25 Chiral chloroalkanes can undergo nucleophilic substitution with NaOH(aq) to form alcohols.

The reaction mechanism is either S_N1 or S_N2.

Which statements are correct?

1 Inversion of configuration occurs in the S_N2 mechanism.

2 Racemisation can occur in the S_N1 mechanism.

3 The hydrolysis of tertiary chloroalkanes takes place by the S_N1 mechanism because the substituent alkyl groups stabilise the carbocation intermediate.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only